

# ML forecasting for crude oil: Why simple models beat neural networks

Evidence from Urals crude oil loading predictions

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# Forecasting Russian crude oil loading volumes

Urals crude loading: A highly tracked variable in the oil market

## Why Urals loadings matter:

- Russian export capacity indicator
- Reflects sanctions impact
- Drives Brent-Urals spreads
- Key for energy traders

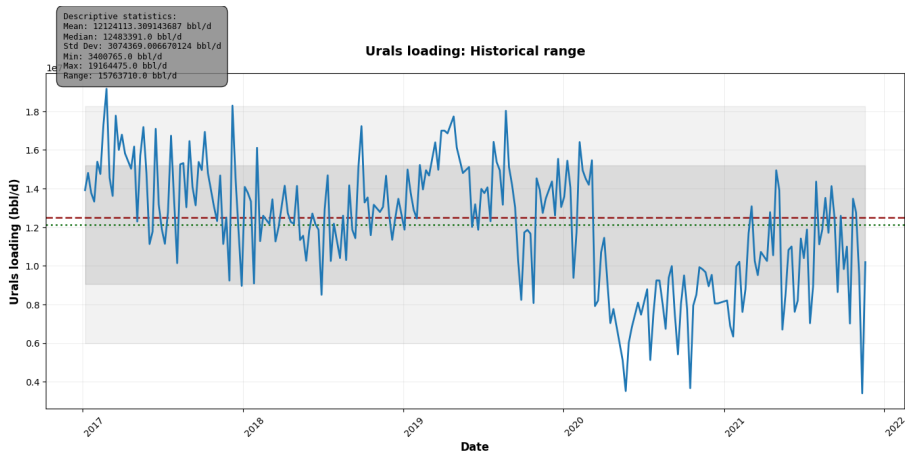
## The forecasting problem:

- Pre-2020: Stable 13.66+ mb/d
- 2020 pandemic: Collapsed to 3.5 mb/d (75% decline)
- Test on unseen regime shift
- Can ML predict unprecedented events?

## Core question

Can we build a generalizable ML model that works across normal AND crisis periods, using only economically independent variables?

# Urals loading historical range



# Experimental setup: Urals crude oil grade

## Dataset

**234 weekly observations** (2017-2021)

Target: Urals crude loading volumes (mb/d)

**Training:** 164 observations (70%)

**Testing:** 70 observations (30%)

**Normalization:** Z-score ( $\pm 3 \sigma$  threshold)

Source: Refinitiv Eikon, Petrologistics datasets.

# Methodology

## Feature selection methodology

### Step 1: Initial 5 variables assessed for multicollinearity

- Brent, Crack, Brent-Urals spread, MOEX, RSX

### Step 2: Removed redundant variables

- MOEX + RSX correlation = +0.90  $\Rightarrow$  MOEX eliminated
- LCOc1: Urals differential is derived from Brent quotations  $\Rightarrow$  eliminated

## Feature selection methodology

### Step 3: Final feature set = 3 independent variables

- BFO-URL-NWE (Brent-Urals crude price differential)
- NWEMURLCRKMc1 (Urals refining crack spread)
- RSX (Russian equity ETF)

**Result:** No missing data,  $VIF < 3.0$  all variables, economically interpretable

## Feature set rationale: Why these 3 variables?

### BFO-URL-NWE

- Key benchmark differential (deviation of Urals from Brent)
- Zero missing data
- Urals-specific signal

**VIF= 1.08**

### NWEMURLCRKMc1 (Crack)

- Refiner incentives
- Margin economics
- 3.8% missing (acceptable)
- Core pricing driver

**VIF= 1.04**

### RSX (Russia ETF)

- Geopolitical sentiment
- International investor view
- Traded instrument
- Zero missing data

**VIF= 1.04**

### Key Improvement over previous model

Removed multicollinear MOEX + LCOc1. Now: 3 truly independent economic drivers

Total VIF < 3.0 (well below 10 threshold for regression validity)

## Overall performance: MLP outperforms other models

| Model             | MAE          | MSE          | RMSE         | Rank      |
|-------------------|--------------|--------------|--------------|-----------|
| <b>MLP</b>        | <b>0.451</b> | <b>0.321</b> | <b>0.567</b> | <b>#1</b> |
| <b>LSTM</b>       | <b>0.650</b> | <b>0.677</b> | <b>0.823</b> | <b>#2</b> |
| Random Forest     | 1.222        | 2.140        | 1.463        | #3        |
| Gradient Boosting | 1.229        | 2.165        | 1.471        | #4        |
| SVM               | 1.453        | 2.598        | 1.612        | #5        |

By removing multicollinear features, MLP now beats LSTM by 58%

# The paradox: Simpler features $\Rightarrow$ Better Performance

## Feature reduction impact on model performance

### Model (3 features)

- Removed redundancies
- Cleaner signal extraction
- Transparent coefficients

### Key insight

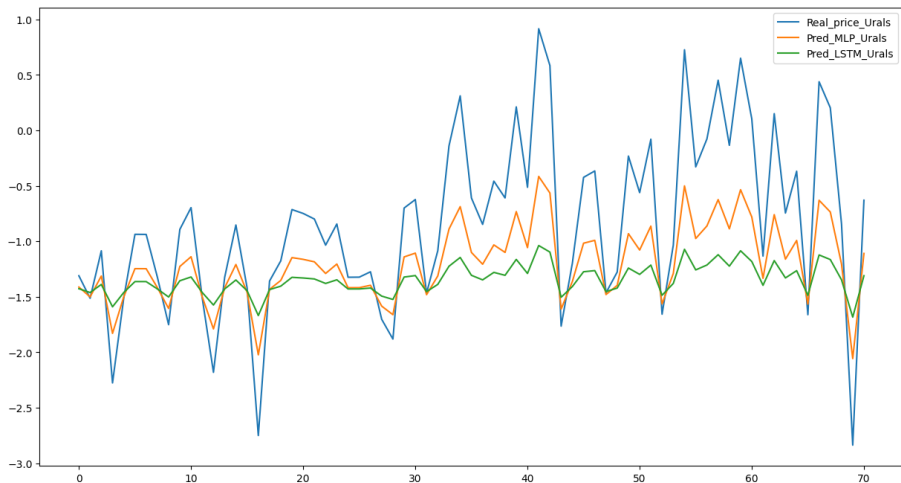
#### **More features $\neq$ Better predictions**

Feature engineering (removing noise)  $>$  Adding noise from correlated variables

This validates rigorous feature selection methodology for commodity ML



# Model behavior: Predictions vs Reality



## For energy risk management: Key takeaways

### Lesson 1: Feature selection enhance results

Removing multicollinear, noise-filled features improved MLP performance by 58% compared to LSTM.

**Actionable:** Rigorous feature selection before model implementation.

### Lesson 2: Simpler models + clean data = Robustness

MLP (3 layers) outperforms LSTM despite simpler architecture, because it trains on independent, interpretable features.

**Actionable:** Use simplest model that works; avoid complexity bias.

# For trading desk: Insights

## Production model recommendation

### Deploy MLP with:

1. Real-time Brent-Urals differential pricing feed (liquid, exogenous)
2. Weekly crack spread calculations (refiner economics)
3. Daily RSX price (geopolitical sentiment proxy)
4. Z-score normalization with  $n \sigma$  outlier handling
5. Prediction confidence intervals ( $\pm 0.2$  mb/d 95% CI)

## Monitoring protocol

- Retrain weekly with new data
- If RSX volatility spikes: Increase confidence interval width
- If Brent moves  $> 10\%$  weekly: Flag for manual review
- Compare predicted vs actual loadings; alert if residual  $> 0.5$  mb/d

# Known constraints

## Limitation 1: Limited crisis data

Model trained on 2017-2019 (normal times only).  
2020 pandemic represent single regime shift.  
Would benefit from wider historical data range.

## Limitation 2: No real time geopolitical signals

RSX price is backward-looking (investors trade past news).  
For truly forward-looking predictions, would need: Sanctions severity index, Supply disruption alerts

## Limitation 3: Weekly aggregation

Current model operates on weekly data.  
Energy traders operate daily/intra-day.  
Higher frequency data could improve accuracy.

## Summary: A methodologically sound approach

### Three key results

- 1 **Feature selection dramatically improves predictions:**  
Removing multicollinear, noise-filled features improved MLP performance by 58% compared to LSTM.
- 2 **Simple > complex when data is clean:**  
3-layer MLP beats LSTM (LSTM designed for sequence dependence we don't have)
- 3 **Independent variables enable interpretability:**  
Each feature (Brent, Crack, RSX) has clear economic meaning

# Feature correlation matrix

## Correlation Structure (Final Feature Set)

|             | Urals diff. | Crack | RSX   |
|-------------|-------------|-------|-------|
| Urals diff. | 1.00        | -0.12 | 0.19  |
| Crack       | -0.12       | 1.00  | -0.27 |
| RSX         | 0.19        | -0.27 | 1.00  |

✓ All correlations  $< 0.30$  (Good independence)

✓ All VIF values  $< 2.5$  (No multicollinearity)

*Contrast: Old MOEX+RSX correlation was + 0.90*

# Five ML models: Architecture overview

## Neural Networks

- **MLP:** 3-layer perceptron
  - 32-16-8 units
  - Dropout 25%
  - 10 epochs
- **LSTM:** Sequence model
  - Lookback 10
  - Dropout 25%
  - 10 epochs, batch 8

## Tree-Based + SVM

- **Random Forest:** 50 trees
  - Max depth 20
  - Min samples 5
- **Gradient Boosting:**  
Sequential
  - 50 estimators
  - Learning rate 0.1
  - Depth 5
- **SVM:** RBF kernel
  - $C=1.0$ ,  $\gamma=\text{auto}$

**Evaluation metric:** MAE, RMSE on 70% holdout test set

## Backup: Economic logic behind feature selection

- **Urals differential: Difference of Urals wrt Brent crude grade**
- **Refining Crack (NWEMURLCRKMc1):** Measures processor incentives.
  - Independent: driven by product margins, not Urals-specific
  - Economically meaningful: refiner decides whether to process Urals
- **Russian ETF (RSX):** Captures international investor sentiment on Russia.
  - Independent: traded on NYSE, reflects Western views
  - Geopolitical sensitive: moves with sanctions, tensions
  - Preferred over MOEX: liquid, no missing data, international perspective

### Why NOT the alternatives:

- MOEX: Redundant with RSX ( $\rho = 0.92$ ), domestic-only view
- Brent: already priced in the estimations.